Cow Wearable — ESP32 Hardware Design & BOM

This document is a practical, shareable hardware design and BOM for a cow-wearable device built around an ESP32 ("cow smartwatch"). It focuses on a neck-collar form factor but notes alternatives (ear tag / leg band). It’s aimed at a pilot for 10–50 cows.

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1) Design goals

- Comfortable, non-invasive collar form factor.
- Low-power operation (weeks → months on battery).
- Reliable sensing: temperature, activity (accelerometer), optional heart-rate/ECG & rumination microphone.
- BLE connectivity to mobile phone (primary prototype).
- Local storage fallback (SPI flash / SD) and OTA-ready firmware.

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2) Core components (high level)

- MCU + radio: **ESP32-WROOM-32** (BLE + Wi-Fi, low cost, lots of libraries).
- 3-axis accelerometer: **ICM-20948** or **MPU-6050** (ICM-20948 preferred for more features and low-noise).
- Temperature sensor: options below (ear-skin vs core).
- **DS18B20** (waterproof digital temp probe) — external probe; good for prototype.
- **MLX90614** (IR non-contact) — measures surface/skin temp.
- Rumen bolus option: separate vendor (not integrated with collar).
- Heart rate (optional): **Polar-like optical PPG** module (difficult on fur) or **ECG electrodes + INA219/AD8232** for chest/neck placement.
- Rumination microphone (optional): small MEMS microphone (SPH0645LM4H) + simple DSP for chew detection.
- Power: **Li-ion rechargeable 18650** or LiPo 3.7V (capacity choice below) + **TP4056** charger or protected Li-ion charger IC + fuel gauge (MAX17055 optional).
- Power regulation: **Buck-boost / LDO** depending on battery and sensor needs (e.g., MCP1700 LDO for 3.3V).
- Storage: **SPI flash** or **microSD card** for large logs.
- PCB: 2–4 layer custom PCB; keep antenna route clear from metal.
- Enclosure: IP66 rated, padded collar mount, venting for temp probe.

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3) Example BOM (prototype) — quantities = 1

Item	Part Suggestion	Purpose	Est. unit cost (USD)
MCU module	ESP32-WROOM-32	MCU + BLE/Wi-Fi	$4–8
Accel/IMU	ICM-20948 module	Activity, orientation	$5–12
Temp sensor	DS18B20 waterproof probe	Skin/temp probe	$1–3
IR Temp sensor	MLX90614 (opt)	Non-contact temp (opt)	$3–8
Microphone	SPH0645 (opt)	Rumination detection	$1–3
ECG amp	AD8232 (opt)	Heart rate/ECG (opt)	$2–4
Battery	1×18650 Li-ion 3400mAh	Power source	$3–6
Charger	TP4056 board + protection	Charging circuit	$1–2
Voltage reg	Buck regulator / LDO	Stable 3.3V	$0.5–2
Storage	microSD socket or SPI flash	Data logging	$0.5–3
PCB (prototype)	2-layer PCB	Assembly	$5–15
Enclosure & strap	Custom resin case + adjustable strap	Weatherproof collar	$6–15
Connectors & misc	Screws, gasket, pads	Assembly	$1–3
Total (approx)			**$33–80**

> Note: costs vary by supplier and volume. Volume production will reduce per-unit cost significantly.

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4) Sensing & sampling recommendations

- **Accelerometer:** sample raw at 25–50 Hz, compute on-device aggregates at 1 min windows (mean, std, step count, tilt, activity spikes). To save power, store aggregates rather than raw continuously.
- **Temperature:** sample every 1–5 minutes. For estrus detection, 5-min resolution is fine. For fever detection, 1–5 minute samples are good.
- **PPG/ECG:** if used, short bursts of 10–30 sec every 10–30 minutes (PPG consumes more).
- **Microphone (rumination):** low-power duty-cycle (e.g., 10s every 60s) or event-triggered recording, then on-device simple DSP to count chew events.
- **Storage:** keep rolling buffer for last 7–14 days locally (aggregates). When BLE connects, sync newest data.

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5) Power budgeting (ballpark)

Assume: ESP32 active during sensor reads + brief BLE sync; deep-sleep between readings. - ESP32 deep-sleep average: **~20–150 µA** (best-case ~20–40 µA with peripherals off).
- Active wake + sensors for 5s every minute: ESP32 active current ~80–200 mA during TX; IMU ~1–3 mA while sensing.

Simple example: - Sample accelerometer + temp every 60s; active time 3s per minute. - Average current ≈ (3s/60s)*200mA + (57s/60s)*0.05mA ≈ 10mA avg. - With 3400 mAh battery: ~3400mAh / 10mA ≈ **340 hours ≈ 14 days**.

If you reduce sampling (e.g., aggregate on device, sample every 5 min) or use LoRa with scheduled transmission, battery life can extend to months. Optimize by: lower active time, reduce BLE TX frequency, push processing to edge, use efficient sensors.

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6) Mechanical & enclosure notes

- Collar should have quick-release and padding.
- Place temperature probe in a small insulated well contacting skin (under hair) or point IR sensor to inside of ear.
- Ensure microphone port is shielded from rain with hydrophobic membrane (e.g., Gore-Tex vent).
- Antenna placement: keep antenna away from metal buckles & battery. Consider an external flexible antenna.

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7) PCB & electronics layout tips

- Keep RF antenna area free from copper planes and large metal objects.
- Route high-current traces (battery to buck) carefully; include an FET cutoff for extreme low-power shutdown.
- Place IMU close to collar mounting point for stable motion measurement.
- Include test pads for UART, SWD/IO for debugging & programming.
- Add a tiny coin-cell RTC or use ESP32 timekeeping with occasional NTP sync; RTC helps with accurate timestamping when offline.

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8) Firmware architecture (high level)

- **Boot / hardware init** → check battery, sensors.
- **Main loop** → sleep → wake_periodically → sample_sensors → compute_aggregates → log_to_flash → decide_if_sync_needed → BLE_advertise_or_connect → sync → deep_sleep.
- **Power modes:** deep-sleep most of the time; use external FET to truly disconnect extra peripherals during deep-sleep.
- **Over-the-air updates:** maintain secure OTA via BLE or Wi-Fi when in range.

- **Data format:** compact binary with timestamps (use epoch ms or ISO strings) and simple TLV per reading type.

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9) BLE profile & mobile sync

- Device advertises a custom BLE service:
- `Service UUID: COW-WEAR-01`

- Characteristics:

  - `0x1001` Device Info (id, firmware)
  - `0x1002` Battery status
  - `0x2001` Latest aggregates (JSON/binary)
  - `0x2002` Bulk sync (MTU-based transfer of logs)
  - `0x3001` Commands (trigger immediate sample, firmware update)

- Sync strategy: mobile app scans for devices, connects, pulls incremental logs, then marks them acknowledged. Use a simple checkpoint (last synced timestamp).

- Security: pair devices or use passkey; encrypt BLE link if sensitive.

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10) Data schema (compact example)

- **AggregateRecord** (per minute): `{id, ts, acc_mean, acc_std, acc_peak_count, temp_mean, rumination_score, battery_mv}`
- **EventRecord**: `{id, ts, type: ['estrus_flag','fever_flag','low_battery'], value, confidence}`

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11) Prototype timeline (suggested)

1. Week 1: order parts + breadboard test (ESP32 + IMU + DS18B20).
2. Week 2: firmware prototype: sampling + BLE advertising + log to SPI flash.
3. Week 3: build simple Android app to connect & pull data (or use nRF Connect for testing).
4. Week 4: assemble 3–5 prototype collars, do one-week field test on 3 cows.
5. Week 5–8: iterate on power optimization, enclosure, placement, and labeling.

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12) Risks & mitigations

- **Battery life too short:** optimize sampling, choose higher capacity or add energy harvesting (solar).
- **Sensor noise / false readings:** per-animal calibration & baseline.
- **Farmer acceptance:** comfortable strap, minimal maintenance, simple charging flow.

- **Water ingress:** use IP66/67 enclosure, seal connectors, conformal coat PCB if needed.

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13) Next steps I can provide immediately

- Full schematic (Eagle/KiCad) for this BOM.
- Reference PCB layout suggestions.
- Starter firmware (ESP-IDF or Arduino) that samples IMU + DS18B20, logs to SPI flash, and serves BLE transfer.
- Simple Android React Native app to sync data.

Tell me which of the above you'd like first (schematic, firmware, mobile app, or test plan), and I will generate it.